How can one nucleotide change in PKLR lead to the premature removal of red cells?

PKLR c.1456C>T, p.(Arg486Trp) (R486W) in red-cell pyruvate kinase

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What did the project ask us to explain?

How can changes in gene/protein sequence and structure explain the molecular basis of disease, and what can gene/protein comparison tell us about the affected biological function?

1. **Cause**
   at gene, protein, structure, and function level
2. **Reference vs disease state**
   how the R486W protein differs from the wild type
3. **Impact on the patient**
   from red-cell metabolism to anaemia
4. **Species comparison**
   human vs rat, mouse, and dog; what position 486 tells us

What is pyruvate kinase deficiency?

An autosomal recessive haemolytic anaemia: red cells are removed prematurely from the circulation, mainly in the spleen.

3.2–8.5

per million
diagnosed prevalence, Western populations (Secrest et al. 2020)

142/177

patients
compound heterozygous (Bianchi et al. 2020)

48%

of 254 patients
with iron overload (Grace et al. 2018)

45%

of 254 patients
with gallstones (Grace et al. 2018)

Treatment: transfusion, splenectomy, iron chelation; the PK activator mitapivat gave a haemoglobin response in 16 of 40 patients vs 0 of 40 on placebo (Al-Samkari et al. 2022).

What does pyruvate kinase normally do?

Phosphoenolpyruvate
+ ADP

pyruvate kinase

last, irreversible step of glycolysis

Pyruvate
+ ATP

Red cells have no mitochondria: they depend on glycolysis for ATP

PKLR
chromosome 1q22
two tissue-specific promoters;
mRNAs differ in their first exon

R-type, 574 residues
red cells (our sequences)

L-type, 543 residues
liver; numbering 31 lower
(R486 = L455)

R-type pyruvate kinase

- homotetramer
- each subunit: N, A, B, and C domains
- allosteric activator fructose 1,6-bisphosphate (FBP) binds in the C domain

Sources: Bianchi and Fermo 2020; Valentini et al. 2002; UniProt P30613.

What changed in the supplied sequences?

Codon / residue	482	483	484	485	486	487	488	489	490
Wild-type CDS	CAG	CTT	CTG	TCT	**CGG**	TAC	CGA	CCT	CGG
Mutant CDS	CAG	CTT	CTG	TCT	**TGG** c.1456	TAC	CGA	CCT	CGG
Wild-type protein	Gln	Leu	Leu	Ser	**Arg**	Tyr	Arg	Pro	Arg
Mutant protein	Gln	Leu	Leu	Ser	**Trp**	Tyr	Arg	Pro	Arg

NM_000298.6:c.1456C>T
the only difference in 1725 nucleotides

p.(Arg486Trp), R486W
the only difference in 574 residues

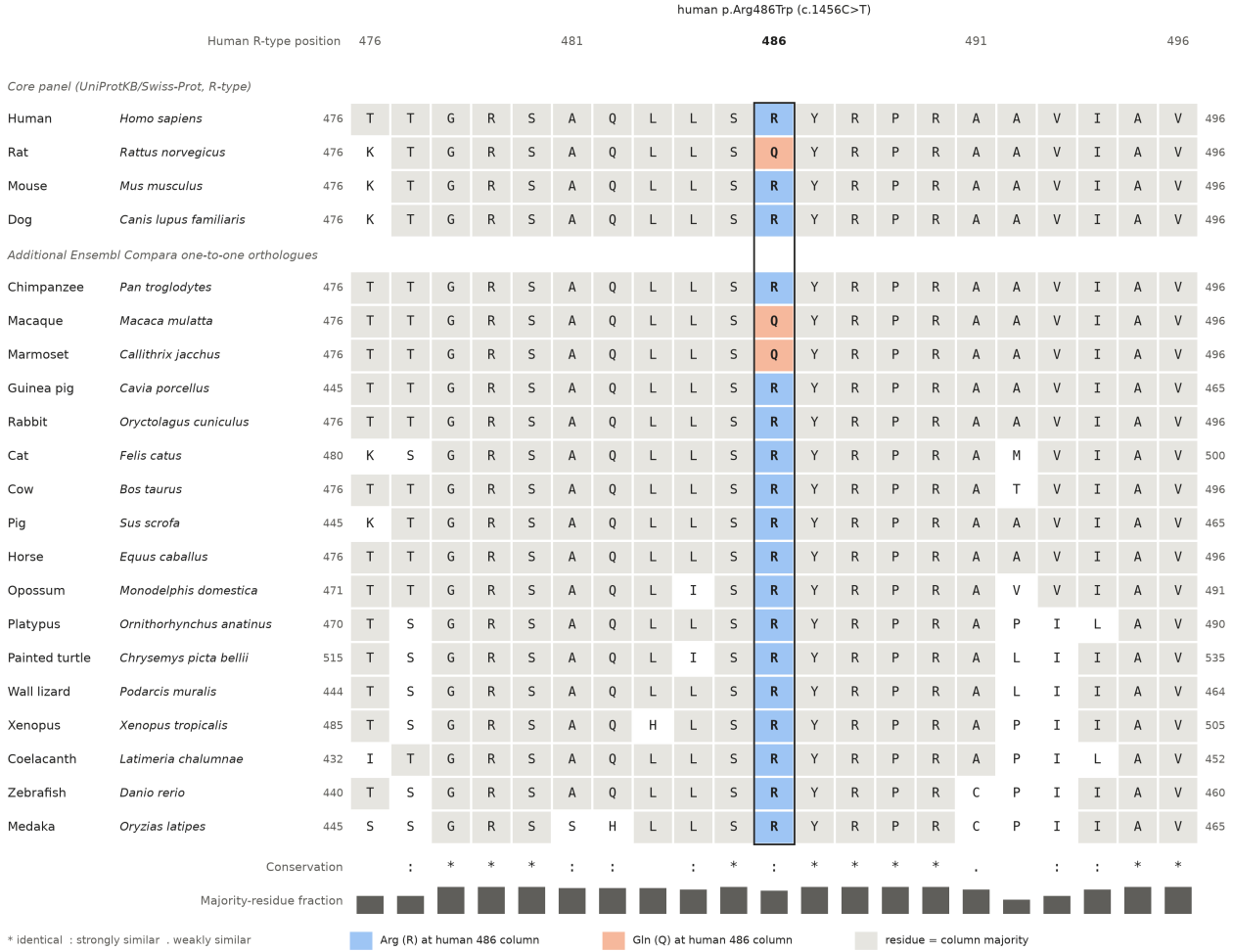
CpG site, exon 10
exon 11 in the 12-exon gene numbering

L-type: c.1363C>T, p.(Arg455Trp)
transcript NM_181871.4

Wild type identical to NM_000298.6 (MANE Select), NP_000289.1, and UniProt P30613-1.

Speaker: Bruno Carrasquilla

Is arginine 486 conserved across species?



MEASURED

19 of 21 columns

identical in human, rat, mouse, and dog (476–496)

MEASURED

Position 486

Arg in human, mouse, and dog; Gln in rat, macaque, and marmoset

MEASURED

Ser485, Tyr487–Arg490

invariant in all 21 vertebrate sequences

Clustal Omega 1.2.4, default parameters; Swiss-Prot R-type sequences (top) and Ensembl 116 orthologues (bottom).

What does the wider species comparison show?

Protein identity with human PKLR (R-type, 574 residues, no gaps)

91.1%
Rat · 523/574

91.6%
Mouse · 526/574

92.5%
Dog · 531/574

vs 64.1–65.3%
identity of the same proteins to human PKM: these are PKLR orthologues

Residue at human position 486

156 / 170
Arg

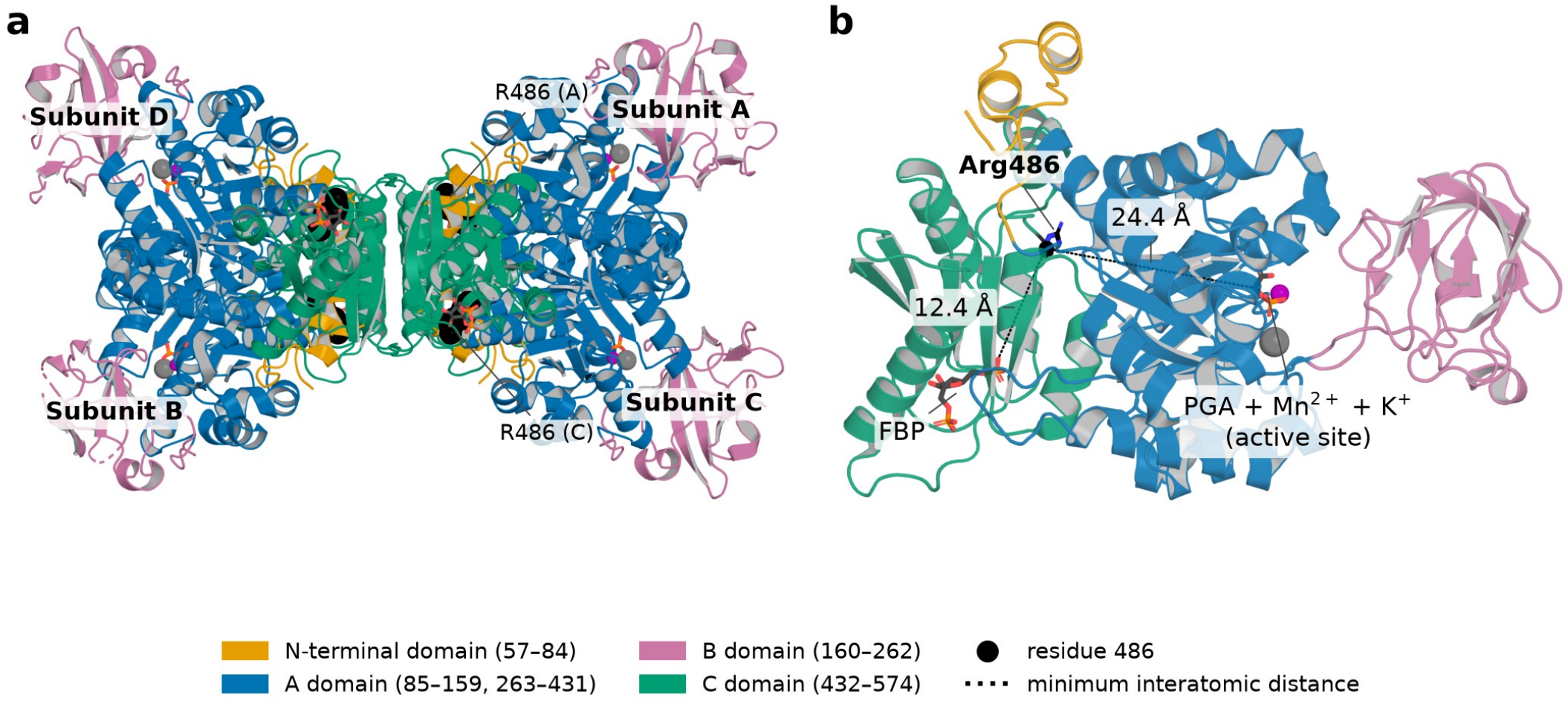
13 / 170
Gln (rat, 10 monkeys, hedgehog, sperm whale)

0 / 190
Trp

INFERRED By maximum parsimony, Arg is ancestral and Gln arose more than once (rat lineage; twice in primates).

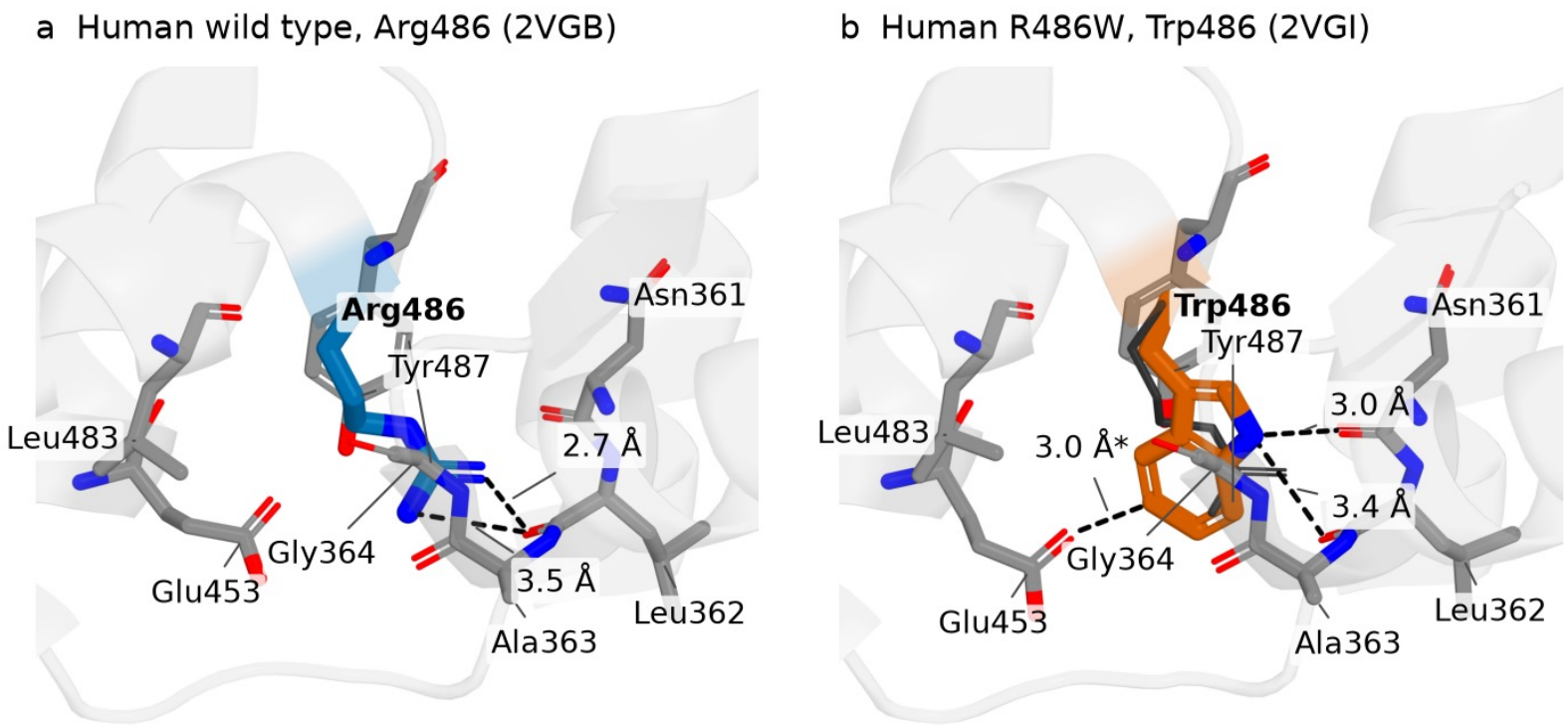
Needleman–Wunsch (EMBOSS Needle defaults); Ensembl 116 Compara homologues; 170 jawed-vertebrate sequences with a resolved residue.

Where is arginine 486 in the enzyme?



PDB 2VGB (wild type, 2.73 Å), subunit A in (b). Relative solvent accessibility of Arg486 0.11–0.12; atoms of other subunits ≥ 6.1 Å away. PGA, phosphoglycolate.

Does tryptophan 486 change the structure?



Arg486 – Leu362 O
2.74–2.90 Å: an A/C domain contact

Trp486 – Asn361/Leu362 O
2.68–3.40 Å: contact kept

Cα RMSD per subunit
0.40–0.59 Å (N, A, C domains 0.21–0.24 Å)

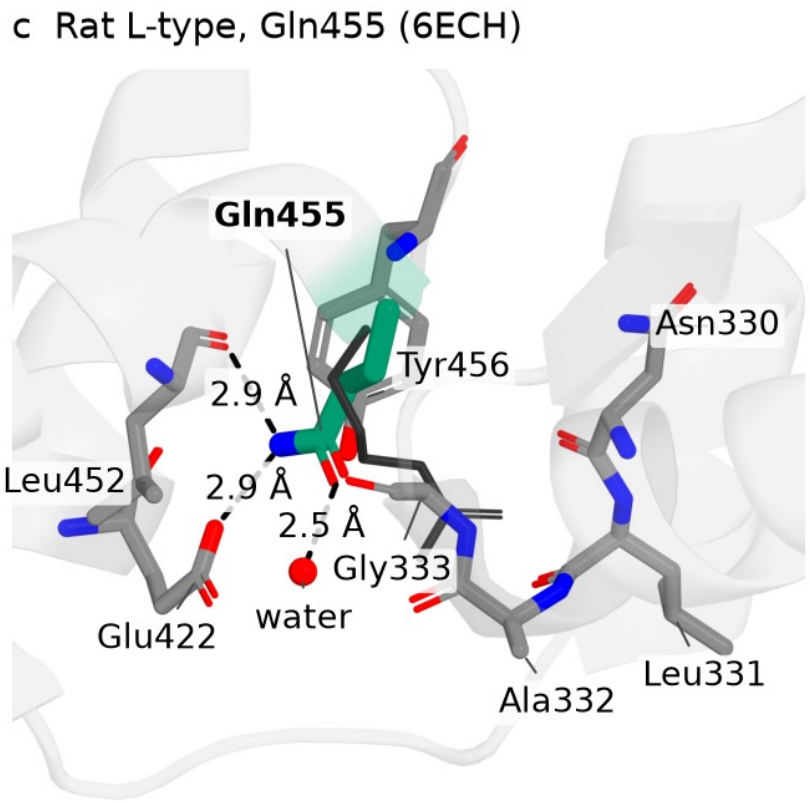
Local Cα RMSD (10 Å)
0.30–0.36 Å, 74th–95th percentile of the whole model

MEASURED Largest local shift: Glu453 OE2 towards Trp486 (1.56–1.75 Å).

PDB 2VGB (wild type, 2.73 Å) and 2VGI (R486W, 2.87 Å); least-squares Cα superposition; ranges over four subunits.

Speaker: Pol Monllau

Does rat glutamine make the same contact?



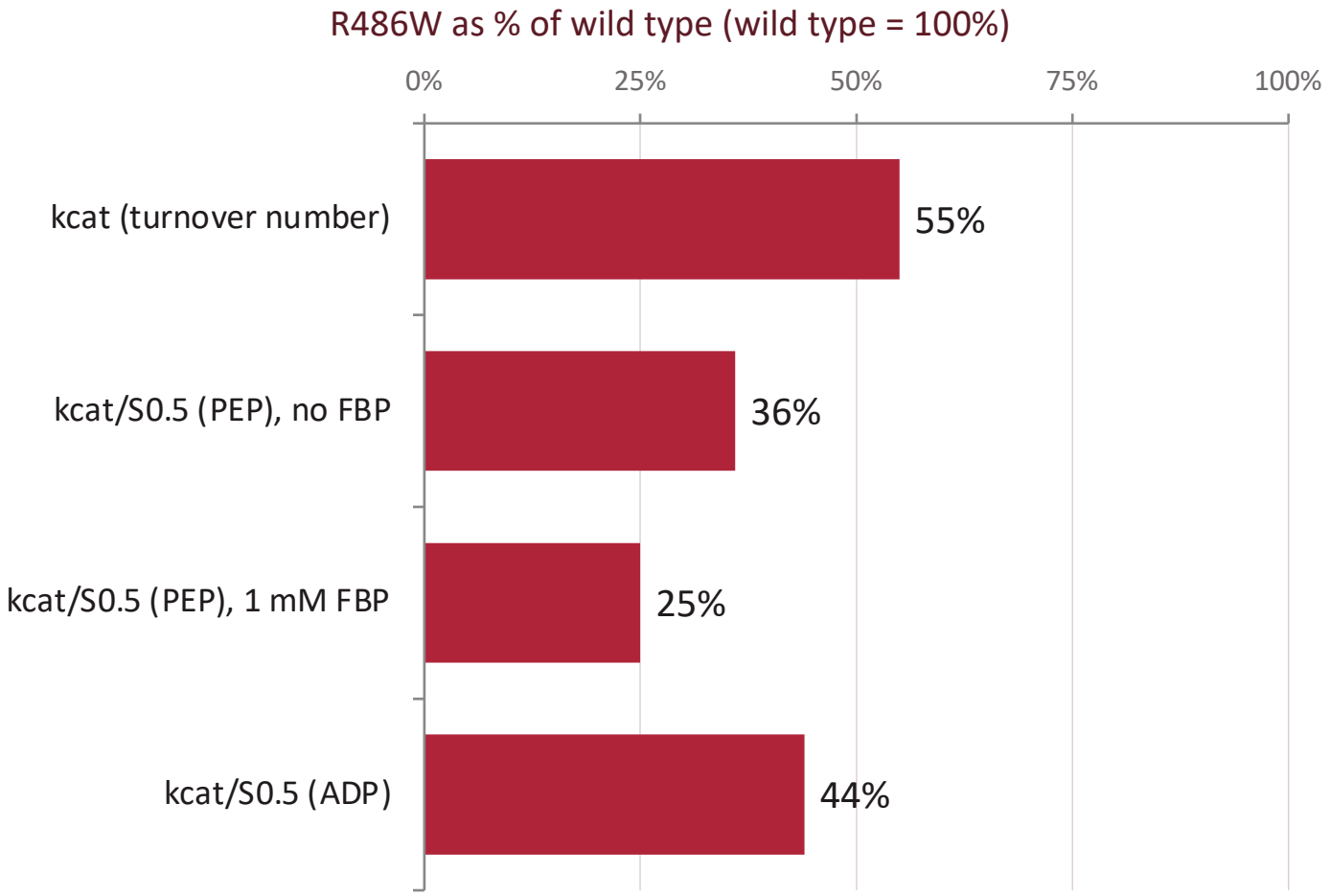
	Human Arg486 (2VGB)	Rat Gln455 (6ECH)
Polar partners ≤ 3.5 Å	Leu362 O (A domain)	Glu422 OE2, Leu452 O (C domain); a water
A-domain carbonyl	2.74–2.90 Å	5.5–6.4 Å: no contact
Distance to FBP	12.4–12.8 Å	11.9–12.0 Å

INFERRED

A non-Arg residue at this position is compatible with a folded, ligand-bound rat tetramer. 6ECH is the phosphorylated L-type (liver) isoform, so it does not show that Gln or Trp is tolerated in human R-type PK.

PDB 6ECH (rat L-type PK, 2.19 Å, phosphorylated at Ser12, with FBP; Gassaway et al. 2019); same contacts in 6ECK. Rat residues = L-type number + 31.

Does the R486W enzyme work differently?



MEASURED

$S_{0.5}$(PEP) 1.10 → 1.69 mM
lower affinity for phosphoenolpyruvate

MEASURED

FBP activation retained
FBP lowers $S_{0.5}$ from 1.69 to 0.40 mM

MEASURED

Thermostability not reduced
at 53 °C, vs wild type

INFERRED

A kinetic defect
not a loss of stability or of allosteric regulation

Valentini et al. 2002, Table II (recombinant enzyme); ratios computed by us. kcat, turnover number; S0.5, substrate concentration giving half-maximal rate.

How strong is the clinical and population evidence?

MEASURED

Pathogenic / Likely pathogenic

ClinVar VCV000001513.102: 33 submitters, no conflicts

MEASURED

9% of PKD alleles

31 of 354 unrelated alleles, second most frequent (Bianchi et al. 2020); most frequent in Italian and Spanish series

MEASURED

0.295%

gnomAD v4.1 allele frequency (4766 / 1 613 440); about one carrier in 170

MEASURED

8 homozygotes

observed in gnomAD; 7.0 expected under Hardy–Weinberg equilibrium

Same codon, c.1457G>A, p.(Arg486Gln): ClinVar Uncertain significance; gnomAD AF 0.0029%, no homozygotes.

Data sources (accessed 22 October 2026): HGNC, NCBI Gene, RefSeq/MANE, Ensembl 116 and Compara, UniProtKB/Swiss-Prot, PDB, ClinVar, gnomAD v4.1 and v2.1.1, OMIM

What do patients' red cells show?

Red-cell PK / hexokinase activity ratio

Healthy controls (mean, n = 15)



11.9

R486W compound heterozygotes (range, n = 4)



1.6–2.7

MEASURED

Whole-blood metabolites, all 15 patients:
phosphoenolpyruvate increased; pyruvate and ATP decreased, the pattern of a block at the PK step

MEASURED

R486W homozygotes
generally mild anaemia

MEASURED

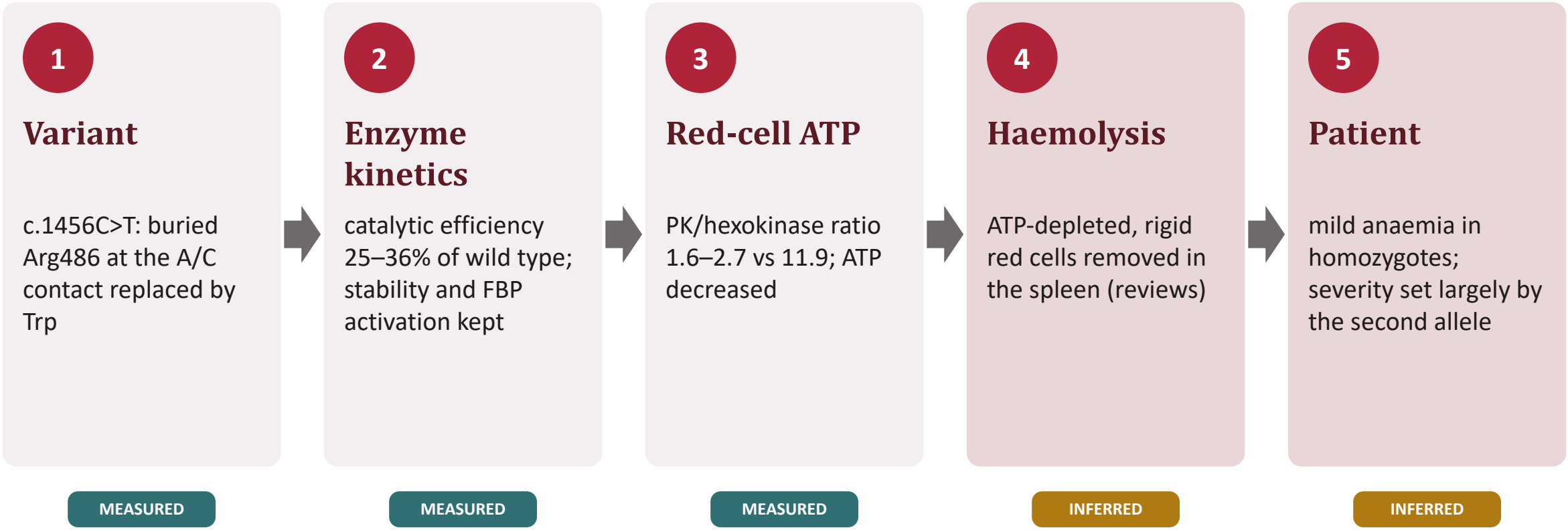
Haemoglobin 9.4–11.9 g/dL
four compound heterozygotes with another missense variant (two splenectomized)

INFERRED

Severity set largely by the second allele
few patients; no series of homozygotes

Valentini et al. 2002 (homozygotes); Rab et al. 2021 (red-cell enzyme activities and metabolites).

How does the variant lead to haemolysis?



Measured links: sequences, structures, recombinant enzyme, patients' red cells. Inferred links: from reviews and few patients.

What is our answer, and what can we not conclude?

What the evidence shows

MEASURED — A CpG transition replaces buried Arg486 at the A/C domain contact by Trp

MEASURED — Fold, active site, and FBP site unchanged; efficiency 25–36% of wild type

INFERRED — R486W is a partial-loss-of-function allele; severity depends on the second allele

INFERRED — Arg486 is ancestral and highly, but not absolutely, conserved; Trp in 0 of 190

What we cannot conclude

UNKNOWN — How a residue 24 Å from the active site lowers catalysis: static, ligand-bound structures cannot show dynamics

UNKNOWN — Whether Gln or Trp is tolerated in human R-type PK: no kinetics of Gln enzymes; rat structure is the L-type

UNKNOWN — Clinical effect in large series: one in vitro study, four compound heterozygotes, no series of homozygotes

How did we build the explanation?

Sequence
identified the variant

Species
showed constraint at 486

Structure
location and contacts; no visible change

Kinetics
what the substitution does

Patients and population
consequence and frequency

No single result explains the disease on its own.

The structure shows where the residue is and what it contacts; the kinetics show what the substitution does; how one leads to the other is not known.

Thank you. Questions welcome.